Motocross Jump Analytics

MAE Task

Setup

A 6-axis IMU (accelerometer and gyroscope) is mounted rigidly to the frame of a motocross bike, near the headstock. The bike is used to record a jumps session with a motocross bike of 2 stroke engine. The session is recorded for 90 seconds.

Dataset :

mx_imu.csv (find in email attachment)
t => time in seconds
accel_x,accel_y,accel_z => acceleration in m/s2
gyro_x,gyro_y,gyro_z => rotational velocity in rad/s
Accelerometer axes: X = forward, Y = left, Z = up.
ODR = 500Hz
Duration = 90 seconds

Objective 1 : Jumps Detection

Implement a jump detector that identifies all the jumps in the dataset using only the accelerometer data.

1. Define your detection criterion mathematically and justify it in 2 sentences.
2. Report the detected takeoff and landing graphs
3. Compute and report air time in seconds for each jump

Objective 2 : FIlter Design

The 2-stroke engine produces strong vibration at ~58 Hz and its harmonic at ~116 Hz on all accelerometer axes.

1. Plot the Power Spectral Density (PSD) of acceleration and gyroscope during a ground segment. Identify and label the engine peaks.
2. Design a Butterworth low-pass filter to remove engine vibration while preserving the motion signal below 20 Hz. Justify your filter order and cutoff.

Objective 3 : Downsampling for power efficiency

Downsample the dataset to identify the lowest sampling rate for the data to identify all the jumps correctly.

1. Identify the lowest sampling rate for which jumps can still be detected and give reasons why.
2. Comment on if any anti aliasing filter is required for the lower sampling rates.